METRIC
MIL-DTL-193B
01 March 2024
SUPERSEDING
MIL-P-193A
28 July 1958
AMENDMENT 1
02 August 1965

DETAIL SPECIFICATION

POTASSIUM SULFATE (FOR ORDNANCE USE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

Reinstated after 01 March 2024 and may be used for new and existing designs and acquisitions.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations, and tests for potassium sulfate intended for ordnance applications.

1.2 Classification. Potassium sulfate furnished under this specification are of the following types:

Type I	for use in the manufacture of propellants
Type II	for use in flash-reducing mixtures
Type III	as a filter in thermosetting resin potting applications
Type IV	for use in ballistic salt rods

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in section 3 or 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to: U.S. Army DEVCOM Armaments Center, ATTN: FCDD-ACE-QSA, Picatinny Arsenal, New Jersey 07806-5000, or emailed to usarmy.pica.devcom-ac.list.ardec-stdzn-branch@mail.mil. Since contact information can change, you may want to verify the currency of this information using ASSIST Online database at https://assist.dla.mil.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916	-	DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://quicksearch.dla.mil/).

2.3 Non-Government standards and publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM E11	-	Standard Specification for Woven Wire Test Sieve Cloth and Test Sieves
ASTM E70	-	Standard Test Method for pH of Aqueous Solutions with the Glass Electrode
ASTM E300	-	Standard Practice for Sampling Industrial Chemicals

(Copies of these documents are available from https://www.astm.org/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Required inspections.

3.1.1 First article inspection. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance. A sample shall be subject to conformance inspection in accordance with 4.3.

3.2 Chemical properties. Potassium sulfate Types I and II shall conform to the chemical requirements specified in Table I.

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TABLE I. Chemical requirements for potassium sulfate.

Requirement	Types I & II
Purity (moisture-free basis) Based on sulfate content, percent minimum Based on potassium content, percent minimum	 99.0 99.0
Moisture, percent maximum	1.0
Insoluble matter, percent maximum	0.1
Grit	None
pH value	7 ± 1
Chlorides (as potassium chloride), percent maximum	0.02

3.3 Granulation requirements. Potassium sulfate shall comply with the granulation requirements specified in Table II.

TABLE II. Granulation requirements.

U.S Alternate Sieve Number	Type I		Type II		Type III		Type IV	
	Min	Max	Min	Max	Min	Max	Min	Max
% Through No.4	-	-	100	-	-	-	-	-
% Retained on No. 30	-	-	30	-	-	-	-	-
% Through No.50	100	-	-	5	-	-	-	-
% Retained on No. 70	-	5	-	-	-	-	-	5
% Through No.100	-	-	-	-	-	-	90	-
% Through No.200	-	-	-	-	100	-	-	-
% Retained on No. 325	-	-	-	-	-	20	-	-

3.4 Average particle size. The average particle size for Type III shall be not less than 0.5 microns.

3.5 Workmanship. Potassium sulfate shall be uniform and free from visible impurities and foreign material such as dirt, rust, paint, wooden particles, or any other foreign objects.

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4. VERIFICATION

TABLE III. Requirements/verification cross reference matrix.

Method of verification 1 – Analysis 2 – Demonstration 3 – Examination 4 – Test						Classes of verification A – First article B – Conformance		
Section 3 Requirement	Description	Verification Methods				Verification Class		Section 4 Verification
		1	2	3	4	A	B	
3.1.1	First article	X		X	X	X		4.2
3.1.2	Conformance	X		X	X		X	4.3
3.2	Purity	X			X	X	X	4.4.2
3.2	Moisture	X			X	X	X	4.4.1
3.2	Insoluble matter	X			X	X	X	4.4.3
3.2	Grit	X			X	X	X	4.4.3
3.2	pH	X			X	X	X	4.4.4
3.2	Chlorides	X			X	X	X	4.4.5
3.3	Granulation	X			X	X	X	4.4.6
3.4	Average particle size	X			X	X	X	4.4.7
3.5	Workmanship			X	X	X	X	4.4.8

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see paragraph 4.2).
b. Conformance inspection (see paragraph 4.3).

4.2 First article inspection. When specified (see 6.2), a sample shall be subjected to first article verification in accordance with Table IV. Samples of not less than 230 grams (g) shall be submitted for examination. Samples shall be produced by the contractor using the same process procedures, material supply sources, production equipment, and production line as will be used in regular production.

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Table IV. First article acceptance tests.

Category	Examination or Test	Requirement Paragraph	Inspection Method
Critical	None defined		
Major			
101	Purity	3.2	4.4.2
102	Moisture	3.2	4.4.1
103	Insoluble matter	3.2	4.4.3
104	Grit	3.2	4.4.3
105	pH	3.2	4.4.4
106	Chlorides	3.2	4.4.5
107	Granulation	3.3	4.4.6
108	Average particle size (Type III only)	3.4	4.4.7
Minor			
201	Workmanship	3.5	4.4.8

4.2.1 Rejection. If any sample fails to comply with any of the applicable requirements the first article quantity shall be rejected.

4.3 Conformance inspection.

4.3.1 Inspection lot formation. Inspection lots and batches shall comply with the MIL-STD-1916, paragraph "Formation and identification of lots or batches." A lot shall consist of a single batch or multiple batches that have been blended to a uniform consistency of potassium sulfate produced by one manufacturer in accordance with one specification and revision under one continuous set of operating conditions. Each batch shall consist of a quantity of potassium sulfate that has been subjected to the same chemical and physical process intended to make the final product homogeneous.

4.3.2 Sampling. Samples shall be collected from the beginning, middle, and end of every batch in every lot in accordance with ASTM E300. A composite sample of not less than 450 g shall be formed which is assumed to be representative of that batch. All composite samples shall be verified in accordance with 4.4. Failure of a sample to comply with any of the requirements specified herein requires rejection of the batch represented by the sample.

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4.3.3 Potassium sulfate.				
Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Purity	4.3.2	3.2	4.4.2
102	Moisture	4.3.2	3.2	4.4.1
103	Insoluble matter	4.3.2	3.2	4.4.3
104	Grit	4.3.2	3.2	4.4.3
105	pH	4.3.2	3.2	4.4.4
106	Chlorides	4.3.2	3.2	4.4.5
107	Granulation	4.3.2	3.3	4.4.6
108	Average particle size (Type III only)	4.3.2	3.4	4.4.7
Minor				
201	Workmanship	4.3.2	3.5	4.4.8

4.3.4 Rejection. If any sample fails to comply with any applicable requirements the batch shall be rejected.

4.4 Test procedures.

4.4.1 Determination of moisture. An accurately weighed sample of approximately 5.0 g of the specimen shall be transferred to a tared moisture dish and heated at 100° to 105°C for 2 hours. The tared moisture dish and contents shall be cooled in a desiccator and weighed. The loss in weight shall be calculated to percentage of moisture in the sample.

4.4.2 Determination of potassium sulfate.

4.4.2.1 Based on sulfate content.

4.4.2.1.1 Preparation of the ion-exchange column. Into a 100 milliliters (mL) beaker containing 50 mL of distilled water add 50 g of Dowex 50W-X8, H + (20-50 mesh) cation exchange resin and allowed to stand for 1 hour until the swelling of resin particles ceases. Transfer the resin to a chromatographic tube 3 centimeter (cm) inner diameter x 38 cm length provided with a perforated plate and teflon stopcock. The resin bed should rest on a pad of glass wool supported by the perforated plate and be 8 cm high. If the resin is in the sodium form, it shall be converted to the acid form by passing 500 mL of 6 molar (M) hydrochloric acid through the column and then washing with distilled water until the effluent shows a pH of 5 or higher. A 0.5 g specimen of potassium sulfate shall be dissolved in 50 mL of distilled water and transferred quantitatively to the column at a rate not exceeding 35 mL per minute. While draining the solution from the column, the liquid level should never be allowed to fall below the top of the resin bed. The column shall then be washed with 4, 50 mL portions of distilled water. The

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column is then ready for use. The resin shall be kept covered with water when the column is not is use. Thus prepared, the column can be used in the analysis of the 7, 0.5 g samples of potassium sulfate before regeneration is necessary.

4.4.2.1.2 Regeneration of the ion-exchange columns. To regenerate the ion-exchange resin pass 50 mL of 6 M hydrochloric acid through the column and follow with distilled water washings until the pH of the effluent water is 5 or higher. Remove the ion-exchange resin from the column and slurry with distilled water. Replace the resin in the column. Dissolve 0.5 g of potassium sulfate in 50 mL of distilled water and pass the solution through the ion-exchange column at a rate not exceeding 35 mL per minute. Wash the column with 4, 50 mL portions of distilled water taking care to stop the flow of the final wash while the resin is still covered with water. The regenerated column can be used in the analysis of 7 additional 0.5 g samples of potassium sulfate before regeneration is again necessary.

4.4.2.1.3 Procedure. Accurately weigh 0.5 g of potassium sulfate to the nearest 0.0001 g and transfer to a clean 250 mL beaker. Add 50 mL of distilled water and swirl gently until all of the sample has dissolved. Transfer the potassium sulfate solution to the resin column and allow it to pass through the column, at a rate not exceeding 35 mL per minute, and into a clean 600 mL beaker. Rinse down the sides of the 250 mL beaker with 50 mL of distilled water. Transfer the rinse water to the resin column when the last portion of the potassium sulfate solution has almost reached the upper level of the resin column. Rinse the column with 3 additional 50 mL distilled water rinses stopping the flow through the column before the last portion of the final rinse reaches the level of the resin. The distilled water rinses shall also be caught in the 600 mL beaker. Add 1.5 mL of concentrated hydrochloric acid to the solution in the 600 mL beaker and heat to boiling on a hot plate. Add 50 mL of 2 percent barium chloride solution (10 g of barium chloride dihydrate in 500 mL of distilled water) dropwise to the heated solution in the 600 mL beaker, over a period of 5 minutes, with moderate stirring. Allow the precipitate to settle out and add a few more drops of barium chloride solution to the clear solution over the precipitate to see if precipitation has been completed. If precipitation has not been completed, add additional barium chloride solution until no further precipitation occurs. Cover the beaker and contents with a watch glass, place it on a steam bath and allow it to digest for not less than 3 hours. Let the beaker and contents stand overnight. Decant most of the supernatant solution through a #3001 fine porosity porcelain crucible, which has been ignited to constant weight (± 0.2 mg) at 600 ± 50 °C, using a Fisher Filtrator or equal. If any precipitate passes through the filtering crucible it shall be poured back into the crucible repeatedly until the filtrate is clear. After most of the supernatant liquid has passed through the filter, quantitatively transfer the remaining precipitate to the filtering crucible with the aid of a rubber policeman. Wash the precipitate with distilled water until the water washing gives a negative chloride test with silver nitrate solution. Dry the crucible and contents in an oven set at 100 ± 5°C for 30 minutes and then transfer to a muffle furnace set at 600 ± 50°C and ignite for 1 hour. Cool the crucible and contents in a desiccator and weigh to the nearest 0.0001 g. Return the crucible to the furnace for 10 minutes, cool in the desiccator and weigh. Repeat this procedure until 2 consecutive weighings agree within 0.0002g. Calculate the percent potassium sulfate as follows:

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$$\text{Potassium sulfate, percent} = \frac{74.65*A}{W}$$

where:

A = weight of barium sulfate precipitate in g.
W = sample weight in g corrected for moisture.

4.4.2.2 Based on potassium content.

4.4.2.2.1 Preparation of solutions.

4.4.2.2.1.1 0.2 M aluminum chloride. 5 g of aluminum chloride ($AlCl_3 \cdot 6\ H_2O$) shall be dissolved in about 50 mL of distilled water and sufficient distilled water shall be added to give 100 mL of solution.

4.4.2.2.1.2 0.1 N sodium tetraphenylboron. 34 g of sodium tetraphenylboron analytical reagent grade shall be dissolved in 900 mL of distilled water at room temperature and diluted to approximately 1 liter (L). 4 mL of 0.2 M aluminum chloride solution shall be added. The solution shall be allowed to stand until it clears. The solution shall be filtered and stored in a polyethylene bottle. The solution shall be re-filtered before use if any turbidity or suspended matter is present and shall not be used when older than 48 hours.

4.4.2.2.1.3 Saturated potassium tetraphenylboron wash solution. A quantity of potassium tetraphenylboron shall be prepared from reagent grade potassium sulfate using the procedure described in 4.4.2.2.2 except that the precipitate shall be washed with 50 mL of 0.5 percent acetic acid solution. A 0.35 g portion of this potassium tetraphenylboron shall be shaken for 2 hours with 300 mL of distilled water contained in a glass stoppered flask, and the resulting solution filtered through a No.40 Whatman filter paper.

4.4.2.2.2 Procedure. An accurately weighed portion of approximately 0.22 g of potassium sulfate shall be transferred to a 250 mL beaker and dissolved in 100 mL of distilled water. The solution shall be adjusted, if necessary, to a pH value of 5 to 6 with diluted acetic acid or sodium acetate. A few drops of 0.2 M aluminum chloride solution shall be added, and the solution heated to 70°C. Fifty milliliters of 0.1 N sodium tetraphenylboron solution shall be added dropwise, with constant stirring. The suspension shall be cooled to room temperature and allowed to stand for 5 minutes. The solution shall be filtered, and the precipitate collected on a tared, medium porosity glass filtering crucible. The precipitate shall be washed with 5, 10 mL portions of the saturated potassium tetraphenylboron wash solution. The crucible and contents shall be dried at 120 ± 2°C to constant weight, cooled in a desiccator, and weighed. The purity shall be calculated as follows:

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$$\text{Purity, percent} = \frac{24.32*A}{W}$$

where:

A = weight of dried precipitate
W = weight of specimen corrected for moisture

4.4.3 Determination of insoluble matter and grit. A weighed portion of approximately 50 g of potassium sulfate shall be dissolved in approximately 500 mL of hot distilled water. The solution shall be filtered through a tared, fine porosity filtering crucible. The crucible and residue shall be washed with hot water until the washing no longer gives a test for sulfate. The crucible and contents shall be dried at 100° to 105°C for 1 hour, cooled in a desiccator and weighed. The increase in weight shall be calculated as percentage of insoluble matter in the sample. The material retained in the crucible shall be transferred to a smooth glass slide and the material shall be pressed and rubbed between glass slides determining the presence of grit by scratching noise and scratches on the glass slides.

4.4.4 pH value. Weight 5 ± 0.05 g of sample and place in a 250 mL beaker. Add 100 mL of neutral distilled water to the beaker and dissolve the sample. Determine the pH value of this solution in accordance with ASTM E70.

4.4.5 Determination of chlorides (as potassium chloride). A weighed portion of approximately 50 g of potassium sulfate shall be transferred to a beaker, dissolved in 400 mL of distilled water, and the solution filtered. 20 mL of 5 percent nitric acid solution and 5 mL of 5 percent silver nitrate solution shall be added to the filtrate. The solution shall be heated to boiling and agitated vigorously to coagulate the precipitate of silver chloride. Exposure of the precipitate to strong light shall be avoided, as strong light causes the exposed surface to decompose and liberate chlorine. The solution shall be filtered through a tared filtering crucible and the residue washed with 1 percent nitric acid. The crucible and residue shall be dried at 140° to 150°C for 2 hours, then cooled in the desiccator for 1 hour. The crucible should have been previously heated at 140° to 150°C for 1 hour and cooled in a desiccator before taring.

$$\text{Percentage of potassium chloride} = \frac{52.02 * A}{W}$$

where:

A = weight of silver chloride
W = weight of specimen

4.4.6 Determination of granulation. An accurately weighed portion of approximately 100 g of potassium sulfate shall be placed on the specified nest of sieves complying with ASTM E11, properly arranged, and assembled with a bottom pan. Two metal washers, weighing not more than 15 g each, shall be added to the upper sieve to help break up lumps. The nest sieve shall be covered and shaken for 5 minutes by hand or mechanically by means of a shaker geared to produce 285-315 gyrations and 150 ± 10 taps of the striker per minute. If necessary, the material on each sieve, beginning with the top one, shall be brushed lightly with a ½ inch flat bristle brush

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until no further material passes through each sieve. The portions retained on the various sieves shall be weighed and the results shall be calculated to a percentage basis as required.

4.4.7 Determination of average particle size. The average particle size of Type III only shall be determined on a Fisher Sub-Sieve Sizer or equivalent instrument.

4.4.8 Workmanship. Take approximately 100 to 200 g of the material and spread the composition out on a clean white surface. The sample shall be examined by personnel with normal vision or normal corrected vision. The sample shall be uniform and free from lumps, visible impurities, foreign matter, or other defects. Failure to meet the workmanship inspection or during following production processes shall result in batch or lot rejection.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. Potassium sulfate covered by this specification is intended for use in ordnance applications.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type of material required (see 1.2).
c. Packaging requirements (see 5.1).
d. Requirements for First article samples (see 4.2).
e. Provision for submission of first article samples.
f. Requirements for submission of acceptance inspection equipment.

6.3 Definitions.

6.3.1 Batch. A discrete quantity of material that has been manufactured through various unit operations to produce a substantially uniform final product.

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6.4 Technical agency. US Army Combat Capability Development Command Armaments Center (DEVCOM AC) is the technical agency that is responsible for the preparation and revision of the specification. All correspondence to the Technical Agency should be submitted through the contracting officer to: DEVCOM AC, ATTN: FCDD-ACE-QEE, Picatinny Arsenal, NJ 07806-5000.

6.5 Approval of equivalent test methods. Prior approval by the Contracting Officer should be required for use of equivalent test methods and/or instruments. A description of the proposed method and/or instrumentation should be submitted through the Contracting Officer to the Technical Agency (see 6.4). This description should include but not be limited to the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision, and drawings of any special equipment required.

6.6 Submission of alternative conformance inspection provisions. Unless otherwise specified in the contract, proposed alternative conformance inspection provisions should be submitted by the contractor for evaluation by the Technical Agency (See 6.4) responsible for preparation of this specification.

6.7 Incorporated changes. ECP Y23Q2007 provides documentation for noted changes.

6.8 Government limited liability. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

6.9 Subject term (key word) listing.

Propellant

6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army-AR
Navy-OS
Air Force-70

Preparing activity
Army-AR
(Project 6810-2023-006)

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